



1.8

Rational Functions and Zeros

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1

| ECHS Mathematics

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Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: A zero occurs where the simplified numerator is zero and the original denominator is nonzero.

2. Write one precise sentence explaining this principle: Denominator zeros are excluded from the domain.

3. Write one precise sentence explaining this principle: A common factor may create a hole rather than a zero or vertical asymptote.

4. Write one precise sentence explaining this principle: Sign charts use all critical numbers: zeros and excluded inputs.

Core

5. Find the zeros of $R(x) = \frac{(x-4)(x+1)}{(x+1)(x-2)}$.
6. Determine the sign of $\frac{x-3}{(x+2)(x-1)}$ on $(1, 3)$.
7. Can an excluded input also be listed as an x-intercept?
8. State what the formula $R(x) = 0 \iff P(x) = 0$ and $Q(x) \neq 0$ tells you and name any required restriction.
9. State what the formula $\text{Dom}(R) = \{x : Q(x) \neq 0\}$ tells you and name any required restriction.
10. Create a second representation (table, graph, formula, or context) for one central relationship in Rational Functions and Zeros. State its domain and justify one key feature.

Medium

11. For $\frac{(x-3)(x+2)}{(x-1)(x+2)}$, find zeros and excluded inputs and classify any cancellation.
12. For $\frac{x^2-9}{x^2-4}$, find zeros and excluded inputs and classify any cancellation.
13. For $\frac{(2x+1)^2}{(x-4)^3}$, find zeros and excluded inputs and classify any cancellation.

14. For $\frac{x(x-5)}{x^2+1}$, find zeros and excluded inputs and classify any cancellation.

15. For $\frac{(x+1)(x-2)}{(x+1)^2}$, find zeros and excluded inputs and classify any cancellation.

Hard

16. Make a sign chart for $R(x) = \frac{(x-2)(x+1)}{(x-3)(x+4)}$.

17. Find k so $x = 2$ is a zero of $\frac{x^2+kx-6}{x+1}$.

18. Can a rational function have no real zeros and no real excluded inputs? Give an example.

19. Explain the graph behavior at a rational zero of even multiplicity.

Difficult

20. Find all k such that $\frac{x^2-kx+4}{x-2}$ has no zero at $x=2$ and instead has a hole there.

21. Construct a rational function with zeros -1 (odd) and 2 (even), a hole at 4 , and vertical asymptote at -3 .

22. For $R(x) = \frac{x^2-1}{x^2-kx+k-1}$, find k so $x=1$ is a hole and $x=-1$ remains a zero.

Challenge / HOT

23. Prove that a reduced rational function changes sign at a simple real zero unless that point is also excluded.

24. Construct two rational functions with the same zeros and excluded inputs but different signs on at least one interval.

AP-Style Multiple Choice

25. A rational zero must make
(A) denominator zero only
(B) simplified numerator zero and denominator nonzero
(C) both numerator and denominator zero
(D) the asymptote zero
26. A common factor that cancels usually creates
(A) x-intercept
(B) hole
(C) horizontal asymptote
(D) new domain point
27. $\frac{x^2-4}{x-2}$ has x-intercepts
(A) -2 and 2
(B) -2 only
(C) 2 only
(D) none
28. Critical numbers for a rational sign chart include
(A) only zeros
(B) only exclusions
(C) zeros and exclusions
(D) y-intercepts
29. An even-multiplicity zero usually
(A) changes sign
(B) does not change sign
(C) is excluded
(D) creates infinity
30. $\frac{x^2+1}{x-3}$ has how many real zeros?
(A) 0
(B) 1
(C) 2
(D) 3

AP-Style Free Response

31. Analyze $R(x) = \frac{(x-2)^2(x+1)}{(x-4)(x+3)}$. (a) Domain. (b) Zeros and multiplicities. (c) Sign chart. (d) Describe x-axis behavior.

32. Find a least-degree rational function satisfying: zero at 3; hole at -1 with y-coordinate 2; vertical asymptote $x=4$; horizontal asymptote $y=1$.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.